SPINE · NEET-SS / FET / INI-SS

Degenerative cervical myelopathy

1. Definitions and pathology

- **Degenerative cervical myelopathy (DCM)** is the umbrella term for cord compression from degenerative disease: **cervical spondylotic myelopathy (CSM)** (disc-osteophyte complexes, ligamentum flavum buckling/hypertrophy, facet hypertrophy, degenerative spondylolisthesis), **OPLL** (ossification of the posterior longitudinal ligament), **ossification of the ligamentum flavum**, and congenital stenosis. **Commonest cause of spinal cord dysfunction in adults over 55.**
- **Static factors:** congenitally narrow canal (**Torg-Pavlov ratio under 0.8**: canal diameter/vertebral body width; **absolute AP canal diameter under 13 mm** = relative stenosis, **under 10 mm** = absolute), disc-osteophyte bars, OPLL.
- **Dynamic factors: extension** narrows the canal (ligamentum flavum buckles; pincer between posteroinferior body and the lamina below); **flexion** stretches the cord over anterior bars; listhesis on flexion-extension films.
- **Ischaemic/cellular injury:** anterior spinal artery territory compromise, demyelination, apoptosis of oligodendrocytes; grey matter (anterior horn) and corticospinal tract involvement.
- **Commonest levels: C5-C6, then C6-C7** (most mobile).

2. Clinical features

- **Symptoms: clumsy hands** (difficulty with buttons, writing, dropping objects), **numb hands** (non-dermatomal), **gait imbalance** (broad-based, spastic, worse in the dark), neck stiffness/pain (not always), Lhermitte sign (electric shock with neck flexion), late **bladder dysfunction**.

- **Upper motor neurone signs:** hyperreflexia, **Hoffmann sign** (flicking the middle finger DIP produces thumb/index flexion), **inverted brachioradialis (radial) reflex** (tapping the brachioradialis gives finger flexion and reduced brachioradialis contraction: lesion at C5-6), **Babinski**, clonus, spasticity, **scapulohumeral reflex** (C3-4 or above), **pectoralis reflex**.
- **"Myelopathy hand" (Ono): finger escape sign** (little and ring fingers drift into abduction/flexion when the fingers are held extended and adducted; ulnar intrinsic weakness), **grip-and-release test** (normal over 20 times in 10 seconds), intrinsic wasting.
- **Gait:** tandem walking difficulty, **Romberg** positive (dorsal column).
- **Radiculopathy may coexist (myeloradiculopathy).**
- **Differentials: motor neurone disease (ALS)** (no sensory signs, fasciculations including the tongue, EMG), **multiple sclerosis** (brain lesions, optic neuritis, oligoclonal bands), **subacute combined degeneration (B12)**, syringomyelia, **normal pressure hydrocephalus** (gait, incontinence, dementia), tumours, peripheral neuropathy, **carpal tunnel syndrome** (may coexist), hereditary spastic paraparesis, copper deficiency, HIV myelopathy, tropical spastic paraparesis (HTLV-1).

3. Severity scoring

- **Modified Japanese Orthopaedic Association (mJOA) score** (18 points: upper limb motor 5, lower limb motor 7, sensory 3, bladder 3): **mild 15-17, moderate 12-14, severe under 12**.
- **Nurick grade (0-5)** based on gait and employment. Others: Cooper, 30-metre walk test, grip and release, Neck Disability Index.

4. Imaging

- **Radiographs:** spondylosis, alignment (**lordosis vs kyphosis**: affects choice of approach), **Torg-Pavlov ratio**, flexion-extension instability, **OPLL** (visible on lateral).

- **MRI (investigation of choice):** cord compression and **intramedullary signal change**:
  - **T2 hyperintensity alone** may be reversible (oedema).
  - **T1 hypointensity with T2 hyperintensity** indicates **myelomalacia/cavitation** (irreversible): **worse prognosis**.
  - "**Snake-eyes**" (bilateral anterior horn T2 hyperintensity on axial images): poor prognosis.
  - Compression ratio, cross-sectional area of the cord.
- **CT: OPLL** extent and type, **double-layer sign** (suggests **dural ossification**: dural tear/CSF leak risk with anterior removal), facet arthropathy, bone quality; **CT myelography** where MRI is contraindicated.
- **OPLL types: continuous, segmental (commonest), mixed, circumscribed (focal)**. Common in **East Asians** (Japan about 2-4%), associated with **DISH**, diabetes, obesity. **Occupying ratio over 60%** and **K-line negative** (OPLL crosses the K-line: line connecting midpoints of the canal at C2 and C7 on lateral) predict **failure of posterior decompression**: use an anterior approach or posterior fusion.
- **Neurophysiology** (SSEPs, MEPs) in uncertain cases and to exclude ALS/neuropathy.

5. Natural history and treatment decisions

- DCM is **usually progressive in a stepwise fashion**; about 20-60% deteriorate without surgery. **Surgery halts progression and often improves function**; outcome depends on **pre-operative severity and duration of symptoms**.
- **AO Spine / WFNS guideline (Fehlings 2017):**
  - **Moderate and severe DCM (mJOA under 15): surgery recommended.**

  - **Mild DCM (mJOA 15-17): surgery or a supervised trial of structured rehabilitation**; operate if it worsens or fails to improve.
  - **Non-myelopathic patients with cord compression on MRI** but radiculopathy (clinical or electrophysiological): higher risk of developing myelopathy: counsel; surgery or close follow-up.
  - **Asymptomatic cord compression without radiculopathy: no prophylactic surgery**; counsel and follow.
- **Predictors of poor outcome: long duration of symptoms**, **severe pre-operative mJOA**, **older age**, **T1 hypointensity/myelomalacia on MRI**, smoking, **multilevel** disease, gait impairment.

6. Surgical options

Anterior approaches (Smith-Robinson; between the carotid sheath laterally and the trachea/oesophagus medially)

- **ACDF (anterior cervical discectomy and fusion):** for **1-2 level disease** with disc-level compression; restores lordosis; direct removal of disc-osteophyte bars.
- **Anterior corpectomy and fusion (ACCF):** compression behind the vertebral body (retrovertebral), focal OPLL, kyphotic multilevel disease. **Hybrid ACDF-ACCF** constructs.
- **Indications for anterior surgery: kyphotic alignment**, **1-3 level** compression, **anterior pathology** (disc, focal OPLL), **K-line negative OPLL**.
- **Complications: dysphagia** (commonest, usually transient; more with multilevel surgery), **recurrent laryngeal nerve palsy** (hoarseness; classically **right-sided** approach (longer variable course of the right RLN) though evidence is mixed), **oesophageal perforation** (rare, serious), **vertebral artery injury** (lateral dissection beyond the uncovertebral joints), **C5 palsy**, **pseudarthrosis** (higher with **multilevel ACDF** and corpectomies),

graft dislodgement (**multilevel corpectomy: add posterior fixation**), **adjacent segment degeneration** (about 2-3% per year, Hilibrand), airway compromise from haematoma (**emergency evacuation** at the bedside), CSF leak (OPLL).

Posterior approaches

- **Laminoplasty** (open-door (**Hirabayashi**), French-door (**Kurokawa**)): **multilevel (3 or more) disease** with **preserved lordosis**; preserves motion; **contraindicated in kyphosis** (cord cannot drift back) and in instability. **Complications: axial neck pain**, **loss of range of motion**, **C5 palsy** (about 5-10%).
- **Laminectomy and fusion (lateral mass screws ± C7/T1 pedicle screws):** multilevel disease with **lordosis or neutral alignment**, instability, axial neck pain; **laminectomy alone is avoided** (**post-laminectomy kyphosis**, especially in young patients and in children).
- **Posterior surgery relies on dorsal drift of the cord**, which needs **lordotic (or at least neutral) alignment** (**cervical lordosis over about 10 degrees**, or **K-line positive** in OPLL).

Approach choice summary

Situation	Preferred approach
1-2 levels, anterior compression	**Anterior (ACDF)**
Kyphosis, focal retrovertebral compression	**Anterior (corpectomy)** or combined
3 or more levels, lordotic, no neck pain	**Laminoplasty**
3 or more levels, lordotic/neutral, instability or neck pain	**Laminectomy + fusion**
OPLL K-line negative / occupying ratio over 60%	**Anterior** or posterior fusion
Multilevel with kyphosis	**Combined anterior-posterior**

C5 palsy

- **Deltoid (± biceps) weakness** appearing **hours to days after surgery** (anterior or posterior), about **5%** (higher after laminoplasty/laminectomy-fusion and corpectomy); proposed mechanisms: **tethering of the C5 root with dorsal cord drift**, foraminal stenosis, reperfusion injury. **Most recover within 6-12 months**; prophylactic **C4-5 foraminotomy** debated.

7. Cervical radiculopathy (related)

- Commonest **C6 (C5-C6 disc)** and **C7 (C6-C7 disc)**. Root = number of the lower vertebra (C5-C6 disc compresses C6 root).

Root	Motor	Reflex	Sensory
C5	Deltoid, biceps	**Biceps**	Lateral arm
C6	**Wrist extensors**, biceps	**Brachioradialis**	Thumb, index, radial forearm
C7	**Triceps**, wrist flexors, finger extensors	**Triceps**	**Middle finger**
C8	Finger flexors	None	Little finger, ulnar forearm
T1	Hand intrinsics	None	Medial arm

- **Spurling test** (extension + rotation + axial compression reproduces radicular pain), **shoulder abduction relief sign** (Bakody).
- **Natural history is favourable**: about 75-90% improve without surgery (analgesia, physiotherapy, selective nerve root injections). **Surgery** for progressive/significant motor deficit or intractable pain after about 6-12 weeks: **ACDF**, **cervical disc arthroplasty** (motion preserving; 1-2 levels without facet arthropathy, instability or OPLL; similar or better outcomes and lower adjacent-level reoperation in

trials), or **posterior foraminotomy** (foraminal soft disc or osteophyte, no instability).

Exam pearl: **Hoffmann sign, inverted radial reflex, finger escape sign, broad-based gait = myelopathy. T1 hypointensity + T2 hyperintensity = myelomalacia (poor prognosis). Moderate/severe DCM (mJOA under 15) = surgery.**

Exam pearl: **Laminoplasty needs lordosis (contraindicated in kyphosis); K-line negative OPLL = anterior. C5 palsy: deltoid weakness after decompression, usually recovers. Dysphagia is the commonest complication of anterior cervical surgery.**

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.